

Trustpilot Review Sentiment Analysis Using Machine Learning

Professional NLP / ML Final Report

983 reviews • TF-IDF (5,000 features) • Logistic Regression C=2 • Flask •
Streamlit

Prepared from the supplied completed notebook and its recorded results.

Abstract

This project develops an NLP sentiment-analysis system for a collected sample of Trustpilot reviews from Uber Eats, Temu, Spotify, Canva, Booking.com and PayPal. The workflow cleans review text, creates rating-based sentiment labels, extracts 5,000 TF-IDF features, compares machine-learning models, tunes Logistic Regression and deploys the selected model through Flask and Streamlit.

The final model is Logistic Regression with $C=2$. It achieved 91.37% overall accuracy and a Macro F1 of 0.5845 on the supplied test results. The most important limitation is class imbalance: only 24 of 983 final reviews are Neutral and only 5 Neutral reviews are in the test set. Neutral test recall and F1 are 0.00.

1. Introduction

Customer reviews provide direct evidence about user experience and service quality. Automated sentiment classification can help summarize large volumes of feedback and provide a repeatable analytical workflow.

2. Problem Statement

The system must classify review text as Negative, Neutral or Positive and provide useful company-level analytics, while preserving the exact trained TF-IDF and model artifacts for deployment.

3. Objectives

- Clean review text.
- Create sentiment labels.
- Perform EDA.
- Extract TF-IDF features.
- Compare and tune ML models.
- Select Logistic Regression $C=2$.
- Deploy through Flask and Streamlit.

4. Dataset

Metric	Value
Original reviews	984
Final reviews	983
Original columns	8
Final columns	14
Companies	6

The original fields include Company, Industry, Review Title, Review Text, Rating, Date, Review_ID and Reviewer. Derived fields include Sentiment, Clean_Text, Word_Count and Character_Count.

5. Data Cleaning

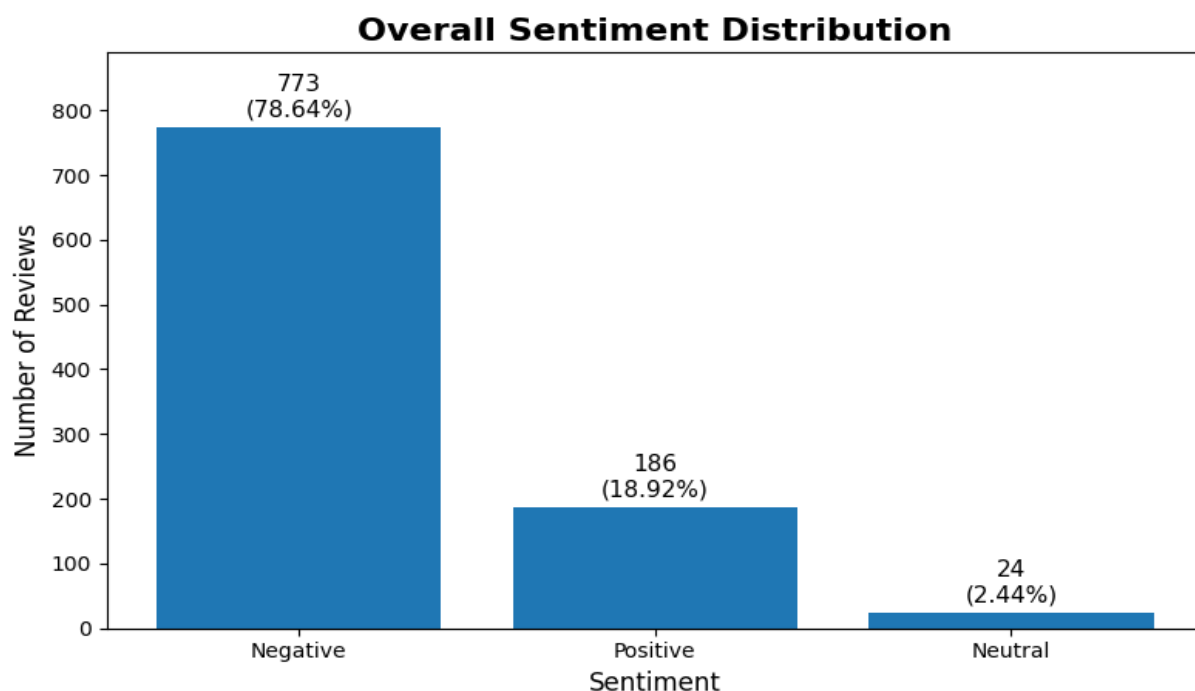
The notebook lowercases text, removes URLs, HTML and email addresses, filters unnecessary characters, normalizes whitespace, removes English stopwords while preserving important negations, and removes one empty/problematic cleaned review. The notebook recorded 184 stopwords.

6. Sentiment Labeling

Rating 1–2 is Negative, rating 3 is Neutral and ratings 4–5 are Positive. Final distribution: 773 Negative (78.64%), 24 Neutral (2.44%) and 186 Positive (18.92%).

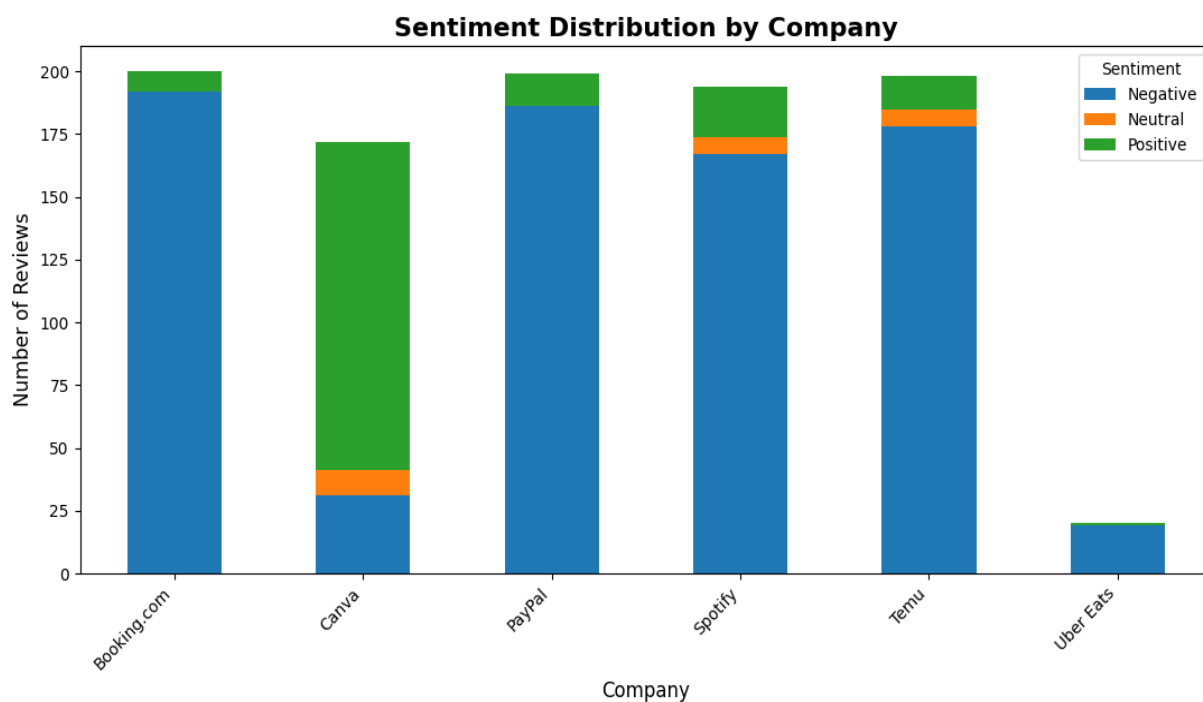
7. Exploratory Data Analysis

Chart 1 — Overall Sentiment Distribution



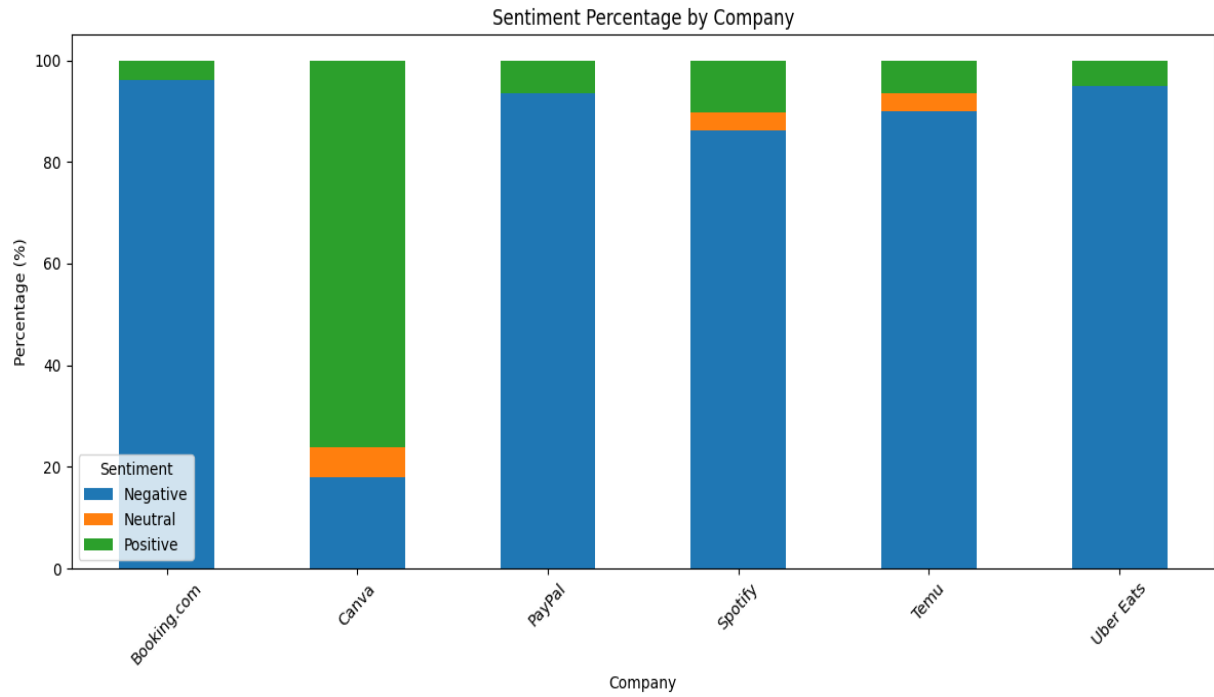
773 Negative, 24 Neutral and 186 Positive reviews. Negative dominates the collected sample. Accuracy is influenced by this imbalance.

Chart 2 — Sentiment Distribution by Company



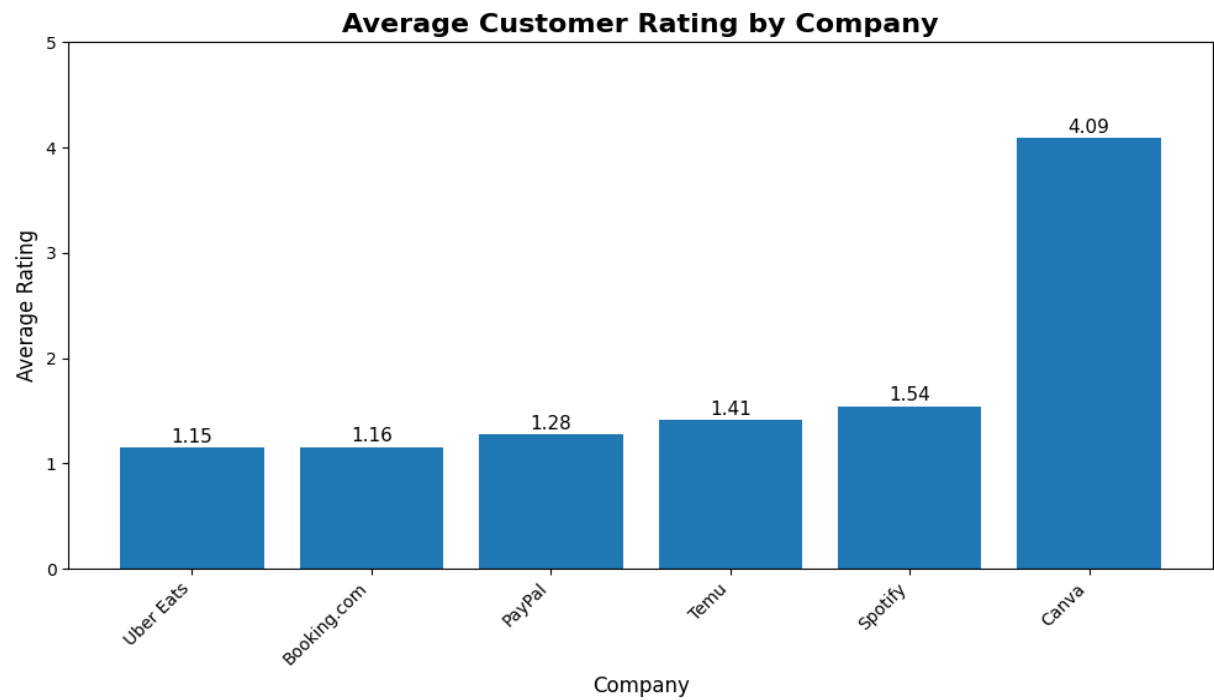
Shows absolute sentiment counts per company. Canva is the positive outlier. Most other company samples are negative-heavy.

Chart 3 — Sentiment Percentage by Company



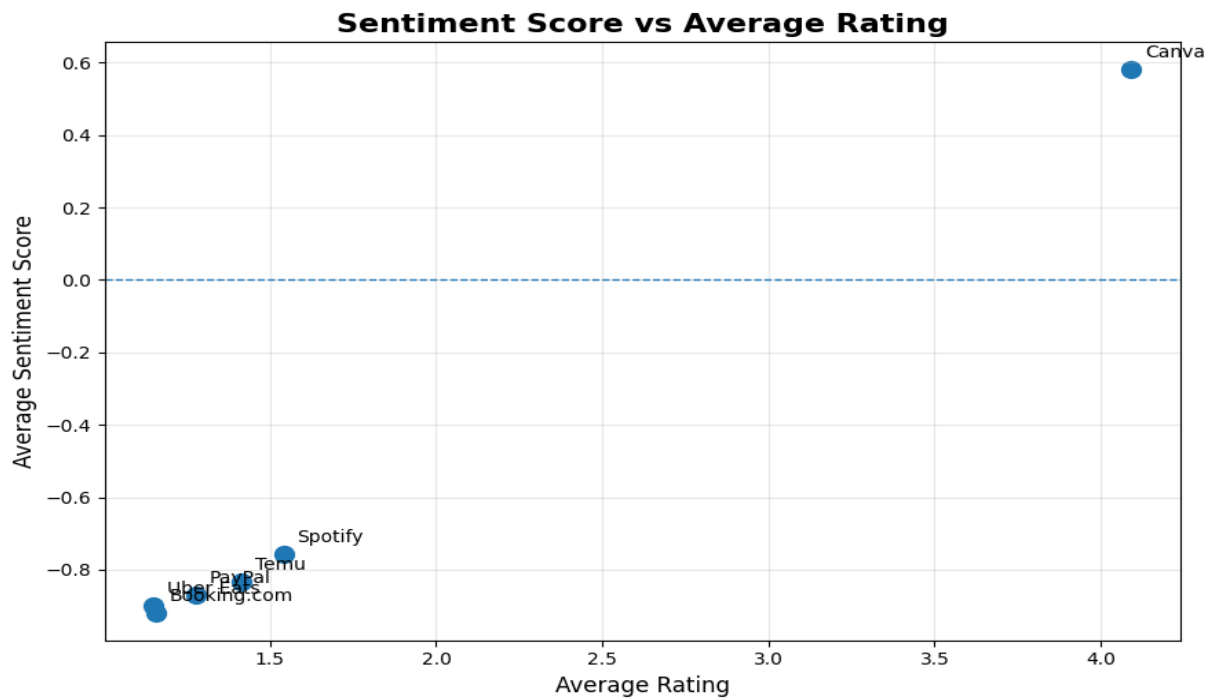
Normalizes each company to 100%. Canva is strongly positive in the collected sample. This is easier to compare across different sample sizes.

Chart 4 — Average Rating by Company



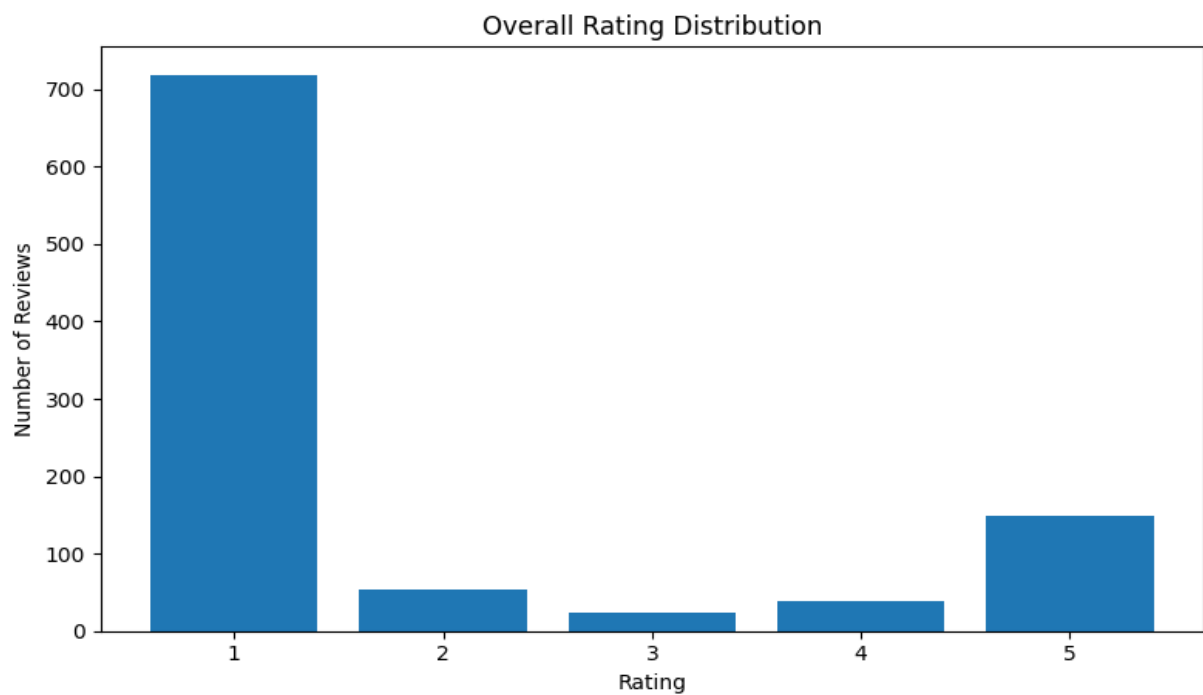
Compares average 1–5 ratings. Canva is about 4.09. Other collected samples are much lower.

Chart 5 — Sentiment Score vs Average Rating



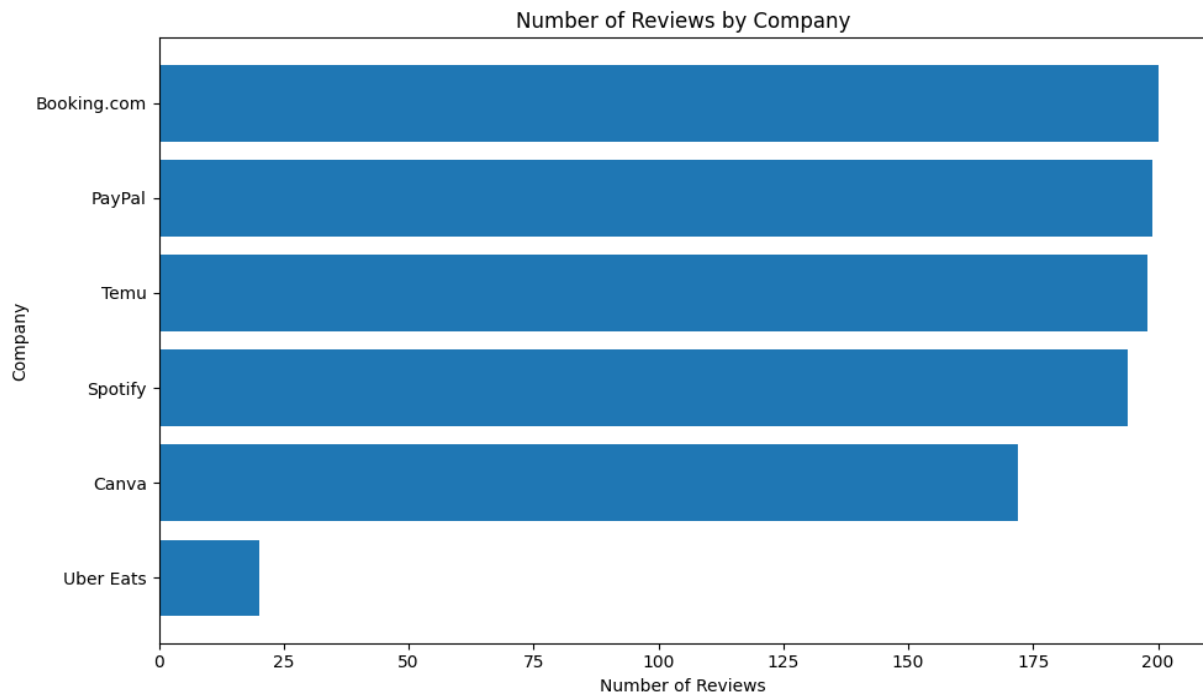
Compares average rating and -1/0/+1 sentiment score. Canva is on the positive side. Other companies are negative in the collected sample.

Chart 6 — Rating Distribution



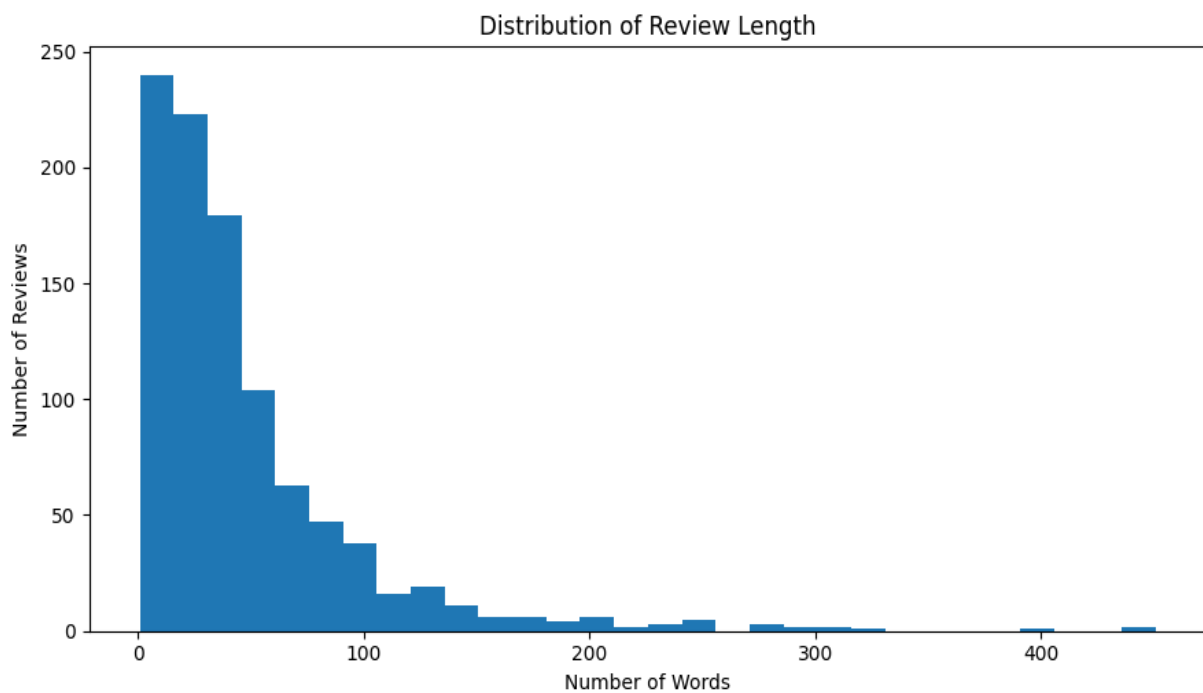
Notebook output: 719 one-star, 54 two-star, 24 three-star, 38 four-star, 148 five-star. One-star reviews dominate. This supports the Negative class imbalance.

Chart 7 — Review Count by Company



Booking.com has 200 reviews and Uber Eats only 20. The sample sizes are not equal. Uber Eats conclusions are less reliable.

Chart 8 — Review Length Distribution



Median word count is 32; maximum is 451. Review lengths vary substantially. Longer reviews may contain more detailed experiences.

8. TF-IDF

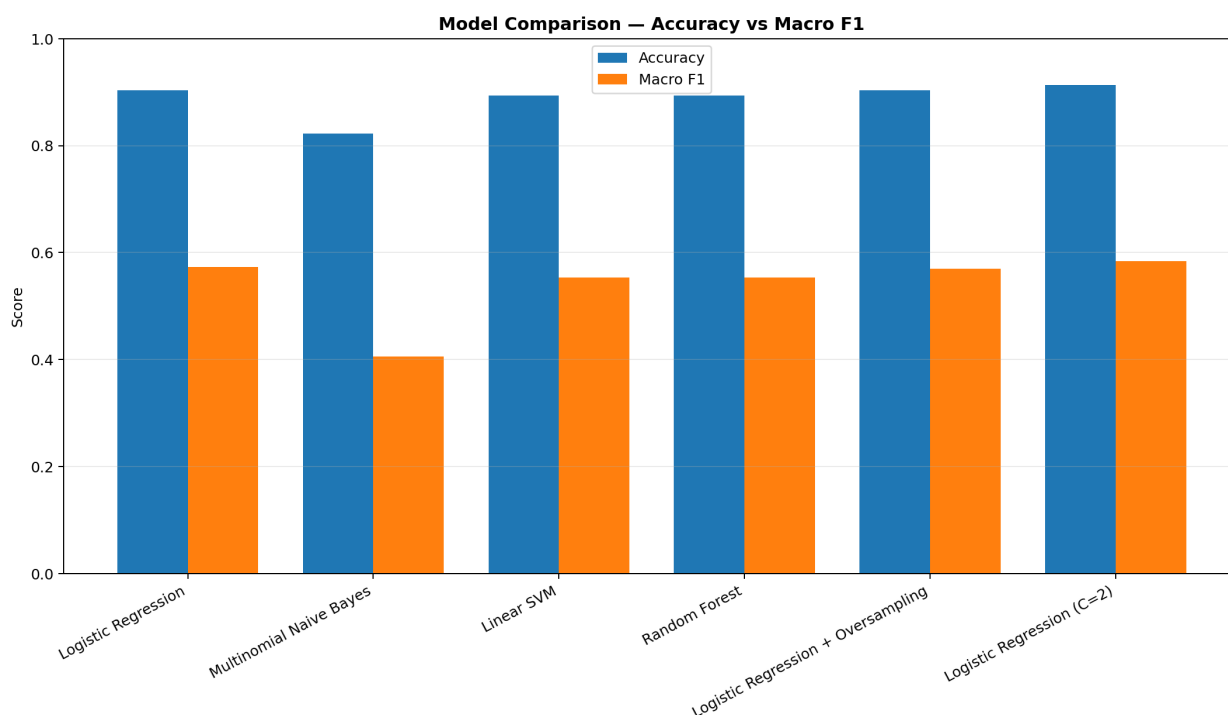
The supplied notebook uses `max_features=5000`, `ngram_range=(1,2)`, `min_df=2`, `max_df=0.95` and `sublinear_tf=True`. Final matrix $X=(983,5000)$. The production application must transform new reviews with the saved fitted vectorizer; it must not fit a new vectorizer.

9. Train/Test Split

Training: 786 samples. Testing: 197 samples. Training distribution: Negative 618, Positive 149, Neutral 19. Testing distribution: Negative 155, Positive 37, Neutral 5.

10. Model Evaluation

Model	Accuracy	Macro F1
Logistic Regression	90.36%	0.5732
Multinomial Naive Bayes	82.23%	0.4056
Linear SVM	89.34%	0.5534
Random Forest	89.34%	0.5536
Logistic Regression + Oversampling	90.36%	0.5699
Logistic Regression (C=2)	91.37%	0.5845



11. Hyperparameter Tuning

C=2 gave the highest Accuracy (0.913706) and Macro F1 (0.584520).

C	Accuracy	Macro F1
0.01	0.873096	0.548522
0.1	0.893401	0.560587
0.5	0.903553	0.572561
1	0.903553	0.573180
2	0.913706	0.584520
5	0.908629	0.577289
10	0.908629	0.577289

12. Final Model

Logistic Regression with C=2. Accuracy = 91.37%. Macro F1 = 0.5845.

Class	Precision	Recall	F1	Support
Negative	0.92	0.99	0.95	155
Neutral	0.00	0.00	0.00	5
Positive	0.93	0.70	0.80	37

Confusion matrix: $\begin{bmatrix} 154 & 0 & 1 \\ 4 & 0 & 1 \\ 10 & 1 & 26 \end{bmatrix}$. The model is strongest on Negative, reasonably effective on Positive, and fails to identify Neutral in the reported test set.

13. Class Imbalance

The 91.37% accuracy must not be interpreted as equal accuracy across all classes. Negative is the majority class, while Neutral is extremely small. Macro F1 and class-level metrics are therefore essential.

14. Feature / Coefficient Analysis

Negative-associated features include: no, not, money, booking, service, customer, paypal, booking com, company, com, account, worst, never, terrible, refund, even, songs, back, payment, horrible.

Positive-associated features include: canva, great, love, excellent, easy, helpful, professional, thank, super, quickly, unbelievable, issue, best, amazing, awesome, perfect, new, wish, projects, excellent customer.

Neutral-associated features include: music, advertise, shirt, fits, everyone, better, fast, free, good, bit high, atrocious, canva free, meaningless, hello, selection, able use, elements, responsive, mentioned, good music.

Coefficients describe feature contribution in the trained model; a word is not universally positive or negative without context.

15. Word Analysis

Chart 9 — Top 20 Words

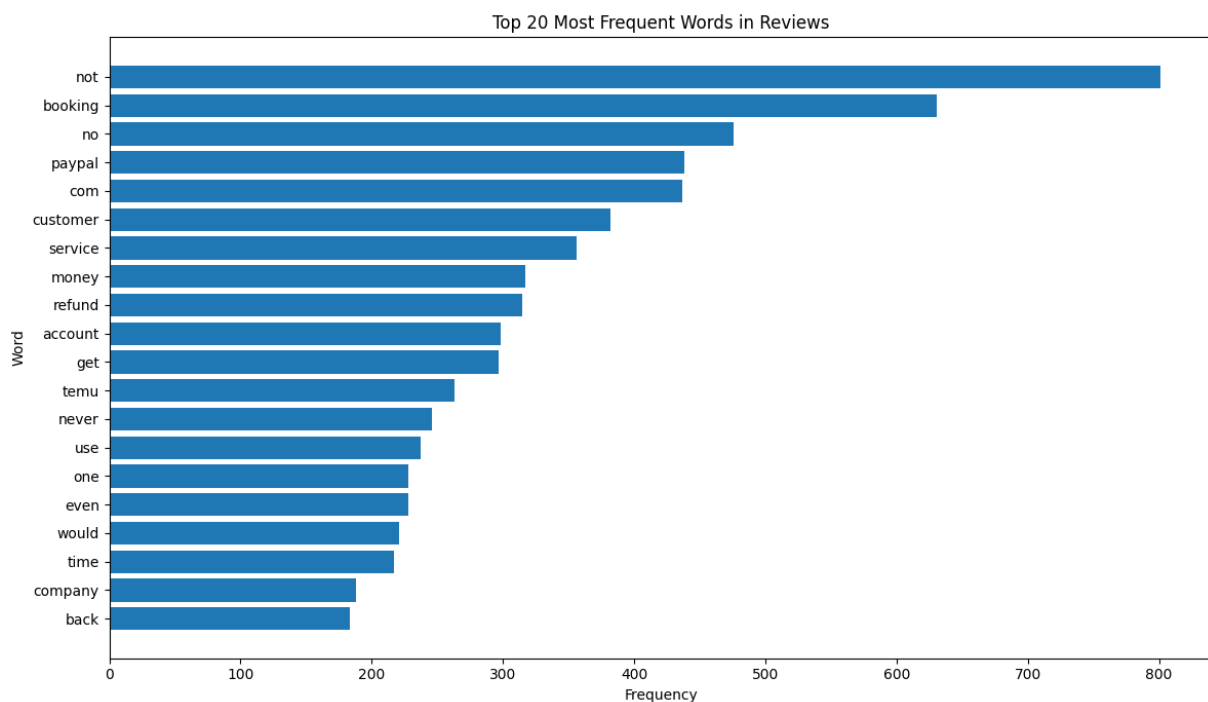


Chart 10 — Top Positive Words

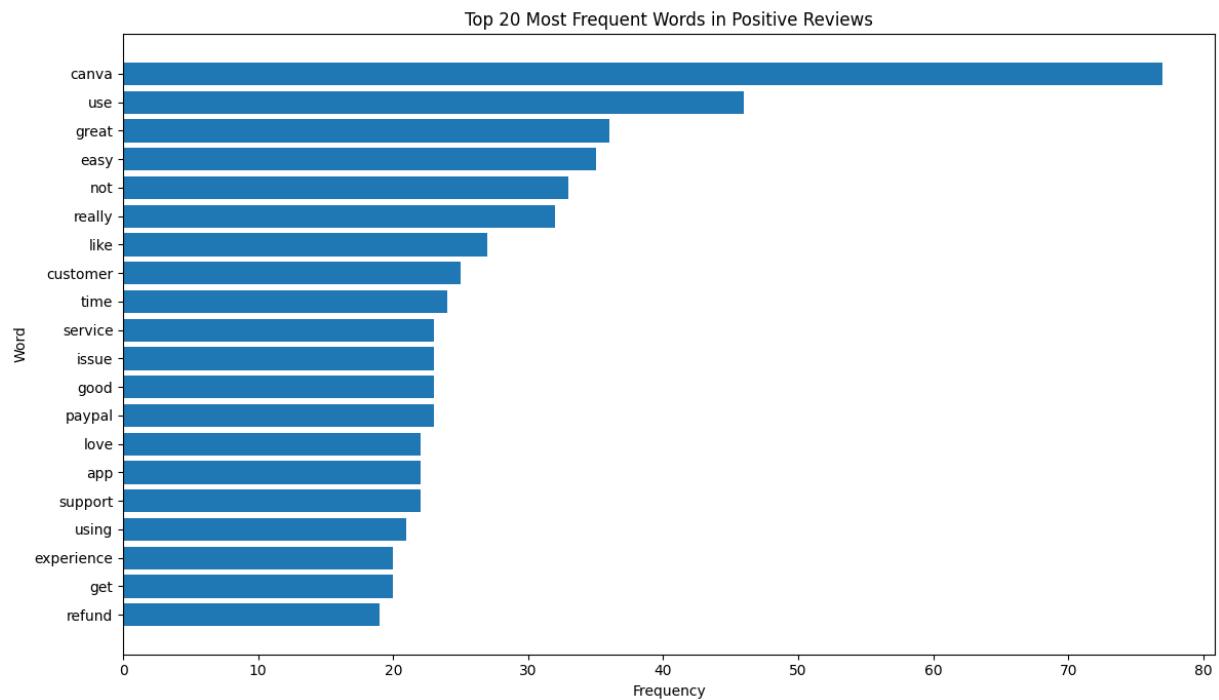
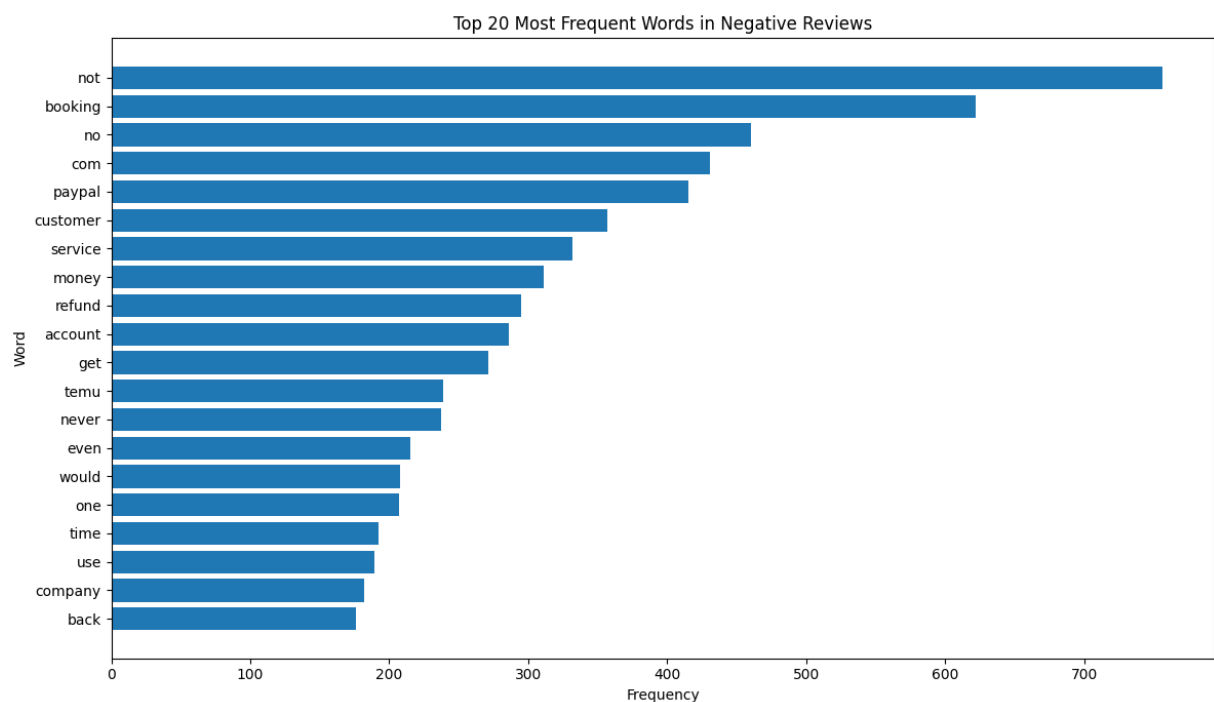


Chart 11 — Top Negative Words



16. Company-Level Analysis

Canva has an average rating of about 4.09 and 76.16% Positive reviews in the collected sample. Booking.com has an average rating of about 1.16 and 96% Negative reviews in the collected sample. PayPal is 93.47% Negative, Temu 89.90%, Spotify 86.08% and Uber Eats 95.00%. Uber Eats has only 20 collected reviews, so that estimate is much less reliable.

These are sample-level findings and should not be stated as each company's complete Trustpilot reputation.

17. Deployment Architecture

Raw review → preprocessing → saved TF-IDF → Logistic Regression C=2 → sentiment + probability → Flask / Streamlit.
One shared prediction pipeline prevents inconsistent logic between applications.

18. Flask Application

The Flask app provides a prediction page, /predict POST route, /api/predict JSON endpoint and /dashboard analytics page. It uses robust file checks and friendly error messages.

19. Streamlit Application

The Streamlit application provides Dashboard, Prediction, Company Analysis, Model Performance, Word Analysis and About Project sections, plus filters, charts, tables and live prediction.

20. Business Insights

Canva is the positive outlier in this sample. Booking.com, PayPal, Temu, Spotify and Uber Eats are predominantly negative in the collected data. These results are useful for the project's analytical demonstration but are not universal reputation claims.

21. Limitations

The dataset is a collected sample; classes are imbalanced; Neutral has only 24 examples; Neutral has only 5 test examples; Uber Eats has only 20 reviews; and TF-IDF is less context-aware than transformer-based NLP.

22. Future Work

Collect more balanced data, improve Neutral coverage, compare transformer models, add multilingual sentiment, explain predictions, monitor drift and automate retraining.

23. Conclusion

The project turns a notebook NLP experiment into a complete deployment-oriented system with charts, documentation, a final presentation, PDF report, Flask app, Streamlit app and a shared production model architecture.

References

Supplied Trustpilot sentiment-analysis notebook and recorded outputs; Trustpilot review pages used during collection; Scikit-learn tooling; Flask documentation; Streamlit documentation.